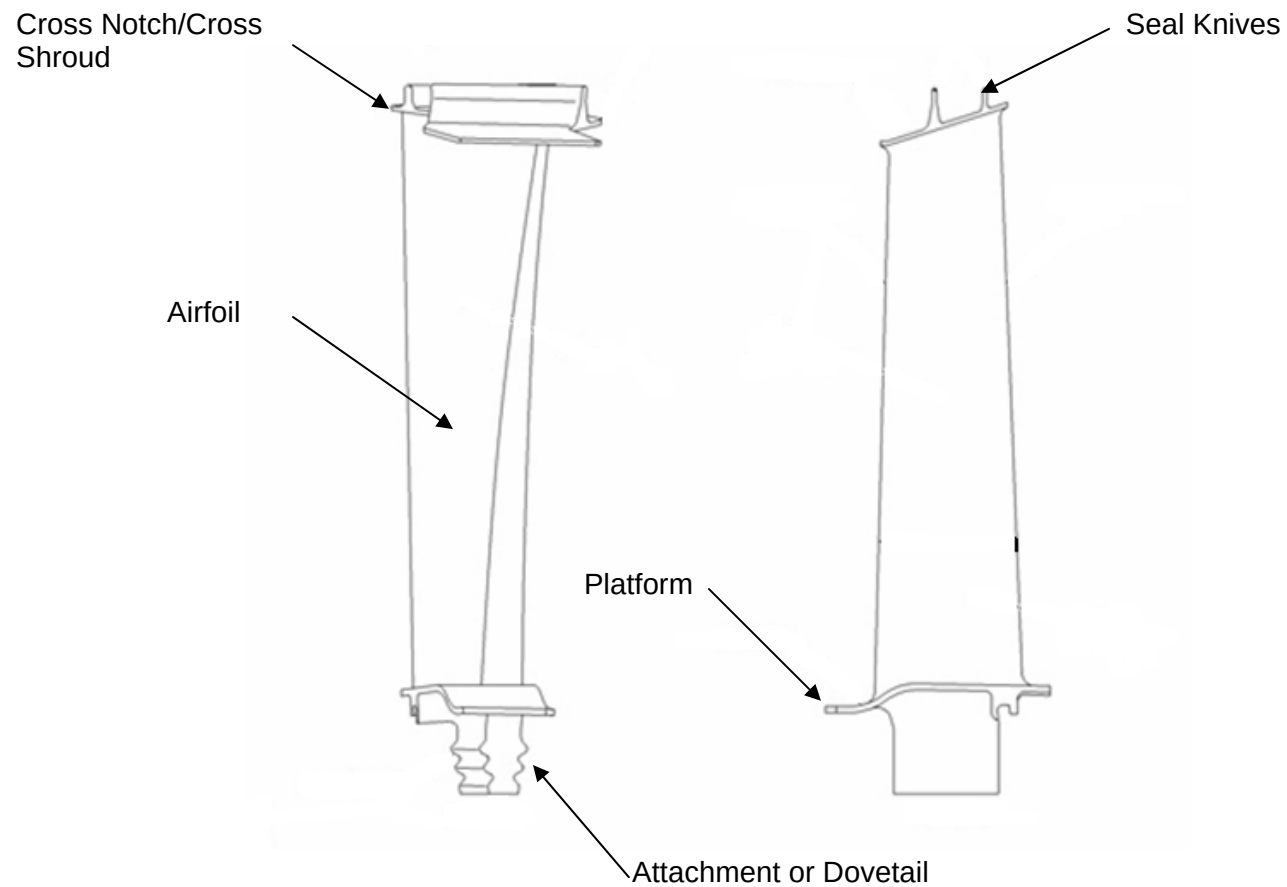


Template #4: Low Pressure Turbine Blade



Nomenclature for Generic Component Features
(all features may not be applicable)

Section 1 – Part Functional capability, i.e., the action(s) that the part is designed to perform in the product				
Section/ Number	Characteristic Critical to Part Functionality	Key Characteristics	Part and System Failure Modes	Risk Mitigation and Control
1-1	Weight	- As-cast surfaces - Machined surfaces	Dovetail/attachment and disk loading-dysfunction	Surface dimensional process control
1-2	Moment of weight - center of gravity - Axial - Tangential - Radial	- As-cast surfaces - Machined surfaces - Mass/density distribution	Dovetail/attachment and disk loading-dysfunction	Surface dimensional process control
1-3	Cooling air utilization (If cooled)	- Total flow rate - Individual circuit-cavity flow rate - Flow distribution - Static pressure distribution - Discharge pressure at showerhead, relative to turbine gas flow pressure	- Low flow: blade over-temperature/dysfunction - High flow: reduced rotor cavity purge/ingestion/part dysfunction - Poor distribution of cooling air: over-temperature, and existing bench flow limits not valid	
1-4	Vibration damping	- Damper fit to blade - Damping ratio	Blade vibration: Blade or disk dovetail cracking	Control damper seating with blade and/or disk
1-5	Vibration characteristics (HCF Capability)	- Blade natural frequencies - Vibration stress distribution - Cycles to failure demonstration - Vibration test set-up and execution criteria, i.e., establishing parameters for blade holding, loading, excitation modes determination, and excitation sensing techniques	Blade resonant vibration, cracking, dysfunction	- Resonance avoidance at steady state operation with margin - Vibratory and steady state stress within Goodman diagram limits
1-6	Creep, LCF, fracture toughness, tensile overload capability	- Blade geometry - Blade cooling - Material mechanical/metallurgical/physical properties - Blade cleaning requirements/effectiveness (internal passages-intergranular attack)	- Blade separation from tensile overload - Crack growth from LCF or creep – rupture - Loss of tip fatigue strength from rubbing	Verify capability exceeds expected utilization

Section 2 – Part Shape, size, dimensions and other physical measurable parameters, and interconnectivity with an engine integral part/system				
Section/ Number	Characteristic Critical to Part Functionality	Key Characteristics	Part and System Failure Modes	Risk Mitigation and Control
2-1	Dovetail or attachment	• Shape • Flatness • Tang relative location and taper • Tolerance (blade root and disk fir tree dimensional spectrums and blade producibility limitations)	• Unbalance tang loading/blade dysfunction • Incorrect loading on disk/cracking	Dimensional process control and inspection
2-2	Platform	• Width • Angel wing length	• Width: binding/damping loss, leakage/cavity ingestion • Angel wing: cavity ingestion or rotor/stator rubs	Dimensional process control
2-3	Airfoil external shape	• Aero contour/stagger angle • Thickness distribution • Twist/lean/bow	• Poor aero performance • Airfoil vibratory mode changes due to thickness distribution • Blade throat area – Bearing load, cavity purge, operability, work split, stage loading	Surface dimensional process control
2-4	Airfoil wall thickness (if cooled)	Thickness vs. chord and span	• Weight • Airfoil stress/life	Dimensional process control
2-5	Airfoil internal cavities (if cooled)	• Passage area and blockage • Turbulator height and shape • Internal wall coating	• Small area: large pressure loss, backflow, ingestion, over-temperature • Large area: Insufficient blade cooling	Dimensional process control and inspection of passage geometry
2-6	Cross notch/cross shroud	Notch shape and thickness • Pre-twist • Seal teeth geometry	• Blade to blade tip loading ◦ Insufficient. Excessive vibe stress, etc ◦ Excessive. High mechanical stress, etc • Poor tip sealing, excessive leakage, poor performance	Dimensional/coating process control
2-7	Tolerance	• Variance-fluctuations in measurements • Measurement gauge repeatability and reliability	Mechanical/Functional, failure due to vibration from nominal	• Verify part is tolerant to variation • Control of manufacturing and inspection processes

Section 3 – Part Materials and Processes – Properties				
Section/ Number	Characteristic Critical to Part Functionality	Key Characteristics	Part and System Failure Modes	Risk Mitigation and Control
3-1	Chemistry	- Major elements - % Variation - Impurities	- Poor mechanical or environmental properties - Poor coating adherence	Material selection Master heat and process control
3-2	Physical properties	- Density - Thermal conductivity - Coefficient of thermal expansion - Refractive index (x-rays) - Melting point	- Excessive blade weight - Excessive thermal stresses - Interference with surrounding hardware	Verify: - Acceptable blade weight - Acceptable stresses - Blade "fit" in hot condition
3-3	Mechanical /metallurgical properties	- Modulus (vs. grain orientation) - Columnar crystal orientation angles - Bare and coated: tensile (UTS, YS, elongation, stress rupture, creep, LCF, HCF) - Long term metallurgical stability - Thin wall effects (if cooled)	- Excessive strain-controlled stress - Undesirable natural frequency - Low material properties, blade dysfunction - Material capability consumed in operation - Low material properties, blade dysfunction	Verify: - Acceptable stresses and frequencies - Grain orientation, as required - Adequate material properties - Operational effects accounted for - Relevant properties are used
3-4	Cast structure	- Grain structure (size, shape, flow, boundaries, gamma prime size and volume fraction) - Low angle boundaries - Directional properties - Hardness - Melting point - Crack propagation rate - Refractive index	Material property reduction due to excessive grain angle or low angle boundaries, blade dysfunction	Verify that grain orientation or low angle boundaries are accounted for Process control and inspection
3-5	Environmental resistance	- Oxidation, corrosion, erosion, fretting resistance - Elevated temperature (creep, diffusion, ageing, temp. gradients) - Rubbing, FOD resistance	Cracking from oxidation/corrosion pitting, blade dysfunction	Verify adequate environmental properties
3-6	Environmental coating	- Coverage and thickness - Long term stability - Oxidation and corrosion resistance	- LCF property reduction due to poor coating selection/requirements, blade cracking - Cracking from oxidation/corrosion pitting	Control of chemistry and application process Verify thermal-mechanical compatibility with base material
3-7	Notch coating	- Coating material composition and density - Coverage and thickness uniformity - Coating and diffusion zone microstructure - Adhesion - Hardness - Residual stress - Stripping requirements - Bonding (interface contamination) - Compatibility with base material/other coatings	High blade stress due to improper blade tip gapping. - Insufficient gap, high mechanical stress, etc - Excessive gap, excessive vibrate stress, etc	Control of chemistry and application process

Section 4 – Part Materials and Processes - Processing				
Section/Number	Characteristic Critical to Part Functionality	Key Characteristics	Part and System Failure Modes	Risk Mitigation and Control
4-1	Casting cleanliness	• Inclusions and porosity • Internal surface cleanliness	Crack growth from inclusion or porosity Poor coating adhesion, corrosion and oxidation	Process control (mold setup, casting process, and filtration) Process control and sequencing
4-2	Heat treatment	• Time @ temperature and ramp rates • Manufacturing sequence • Atmosphere	Incorrect Gamma prime size/spacing leading to low material properties	Process control and sequencing
4-3	Welding and brazing	Effect on base material properties, based on the following: • Weld or braze strength • Microstructure (grain size, shape, flow, braze gap, defects, grain boundary precipitates, gamma prime and volume fraction) • Porosity • Diffusion zone (brazing) • Heat affected zone (welding) • Penetration (welding) • Excess braze allowance • Filler metal selection	Material property reduction	Correct sequencing of braze or weld cycles in the process Process control
4-4	Shot peen	• Intensity • Coverage	Incorrect dovetail compressive stress, cracking, and blade dysfunction	Process control
4-5	Material removal – Non traditional (EDM, ECM, laser, water-jet, etc.) and conventional (machining, milling, broaching, grinding, blending, honing, lapping, stripping, etc.)	Micro cracks or recast: • Inherent to the process • Due to abusive machining Fatigue capability degradation, due to: • Surface finish • Stress risers • Chemical attack	Cracking initiation from micro-cracks or re-cast	Verify part is tolerant to proposed limits Process control
4-6	Non-destructive testing (NDT)	• Acceptance limits • Sensitivity	Cracking from excessive defect size	Verify part is tolerant to allowable defects
4-7	Process sequencing and Significant Process Identification and Substantiation	• Inspection • Manufacturing • Process substantiation and control (RSS, RSA, ESA, etc)	Inadequate final part form, or defect "escape", blade dysfunction	Verify adequate control of processes and parameters

Section 5 – Analysis				
Section/Number	Characteristic Critical to Part Functionality	Key Characteristics	Part and System Failure Modes	Risk Mitigation and Control
5-1	Operating conditions	- Flow path temperatures, pressures, and velocities - Rotor speed	Inadequate part life due to analysis under-predicting part temperature, thermal or mechanical stress	Perform analysis at relevant operating conditions
5-2	Blade cooling (if cooled)	- Secondary flow circuit - Blade cooling flow - Blade cooling	Inadequate part life due to analysis over-predicting the blade cooling	
5-3	LCF life analysis – airfoil and shank	- Temperature effects - Multi-axial effects - Hold time effects - Mean stress effects - Notch effects - Coating effects - Mesh size	Inadequate part life due to analysis - Using incorrect properties - Failing to address the 'key characteristics'	Use applicable properties Select mesh size to correctly model the geometry
5-4	HCF life analysis	- Temperature effects - Vibration and mean stress effects	Blade resonant vibration, cracking, dysfunction	Use applicable properties and Goodman diagram
5-5	Creep and rupture life analysis	- Temperature effects - Material thickness effects	Blade rupture cracking, dysfunction	Use applicable properties
5-6	Fracture mechanics	- Temperature effects - Threshold stress intensity - Crack growth rate	Crack growth from defect	Use applicable dA/dN curve Predict K using expected defect size
5-7	Tolerance to missing coating	- Part temperature - Thermal stress	High thermal stress or part temperature, blade dysfunction	Perform analysis with predicted coating loss
5-8	Variation	- Blade or engine characteristics - Degree of variation required to assure adequate vibratory stress dampening.	Inadequate part life due to vibration	Verify part is tolerant to vibration
5-9	Life calculation	- Predicted stress and Temperature - Root loading profile shifts - Material Properties - Operating during part life - Life impact on disk	Inadequate part life due to analysis under-predicting	Verify correct inputs to lifing process Verify correct 'mission mixing' used

Section 6 – Testing				
Section/Number	Characteristic Critical to Part Functionality	Key Characteristics	Part and System Failure Modes	Risk Mitigation and Control
6-1	Durability	• Fatigue • Corrosion (hot and cold) • Erosion • Creep • Stress rupture • Tensile strength • Cooling effectiveness • Internal contamination • Cleaning requirements • Vibration		
6-2	Over-temperature	• Microstructure • Creep • Hot corrosion • Tensile strength		

Section 7 – Potential Impact on Regulatory Requirements

Note: The regulatory requirements identified below, which are inclusive up to Amendment 20 of 14 CFR part 33, are intended as a guide to applicants when determining the applicable regulations to which they must show compliance. Those requirements listed as “Could be affected” highlight the regulations whose compliance findings are typically affected by the component or part that this template is addressing. This guide is not all-inclusive and the applicant remains responsible for identifying the certification basis of the product on which their PMA part is to be installed.

	Applicable 14 CFR Part 33 Regulatory Requirements		Comments
	Subpart A – General		
1.	33.4 Instructions for Continued Airworthiness	Could be affected	ICAs including on-wing inspection requirements and on-wing limits
2.	33.5 Instruction manual for installing and operating the engine		
3.	33.7 Engine ratings and operating limitations		
4.	33.8 Selection of engine power and thrust ratings		

	Subpart B – Design and Construction; General		
5.	33.14 Start-stop cyclic stress (low cycle fatigue)	Could be affected	Weight and center-of-gravity affect on rotor forces and LCF capability
6.	33.15 Materials	Could be affected	
7.	33.17 Fire prevention		
8.	33.19 Durability	Could be affected	Durability; weight and failure modes relative to containment capability
9.	33.21 Engine cooling		
10.	33.23 Engine mounting attachments and structure		
11.	33.25 Accessory attachments		
12.	33.27 Turbine, compressor, fan, and turbo-supercharger rotors	Could be affected	Weight affect on rotor forces during over-speed
13.	33.28 Electrical and electronic control systems		
14.	33.29 Instrumentation connection		

	Subpart E – Design and Construction: Turbine Aircraft Engines		
15.	33.62 Stress analysis	Could be affected	
16.	33.63 Vibration	Could be affected	Blade vibration-Design
17.	33.65 Surge and stall characteristics	Could be affected	Turbine effective area-Operating line migration, Blade-Rub strip interaction
18.	33.66 Bleed air systems		
19.	33.67 Fuel system		
20.	33.68 Induction system icing (Operability aspects)		
21.	33.69 Ignition system		
22.	33.71 Lubrication system		
23.	33.72 Hydraulic actuating system		
24.	33.73 Power or thrust response	Could be affected	Turbine efficiency/performance
25.	33.74 Continued rotation		
26.	33.75 Safety analysis	Could be affected	Turbine blade release-Case containment

	Subpart E – Design and Construction: Turbine Aircraft Engines		
27.	33.76 Bird ingestion (Operability aspects of ingestion)		
28.	33.77 Foreign object ingestion (Operability aspects of ingestion)		
29.	33.78 Rain and hail ingestion		
30.	33.79 Fuel burning thrust augments		
	Subpart F – Block Tests; Turbine Aircraft Engines		
31.	33.83 Vibration test	Could be affected	Blade vibration
32.	33.85 Calibration tests		
33.	33.87 Endurance test	Could be affected	Blade durability
34.	33.88 Engine over temperature test	Could be affected	Blade over temperature capability
35.	33.89 Operation test		
36.	33.90 Initial maintenance inspection	Could be affected	Blade in-service inspection limits
37.	33.91 Engine component tests		
38.	33.92 Rotor locking tests		
39.	33.93 Teardown inspection		
40.	33.94 Blade containment and rotor unbalance tests (Weight changes)	Could be affected	Failure of the most critical turbine blade while operating at maximum permissible r.p.m.
41.	33.95 Engine-propeller system tests		
42.	33.96 Engine tests in auxiliary power unit (APU) mode		
43.	33.97 Thrust reversers		
44.	33.99 General conduct of block tests		
45.	Part 33 - Appendix A – Instructions for Continued Airworthiness	Could be affected	
46.	Part 33 - Appendix B - Certification Standard Atmospheric Concentrations of Rain and Hail		